

Turboflow Risk Management Status

CTO Brief

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Turboflow Risk Management Status

Executive Summary

Turboflow now has a central Risk Management Policy Wiki in HQ. The wiki gives a governance view across product policy, implementation status, evidence, and gaps.

Current overall status:

Domain	Policy Status	Implementation Status	Highest Risk
Prediction market	Draft from v3.0 spec	Partially implemented	High
CEX pair risk controls	Draft from branch rules and code	Implemented with doc drift	Medium
Risk data inputs	Indexed	Partially implemented / evidence linked	Medium
Admin/runtime config governance	Draft placeholder	Unknown	Medium

The biggest CTO decisions are:

1. Whether prediction-market volatility must be independent per duration, as v3.0 specifies, or whether the current 30s shared volatility cache is acceptable.
2. Whether prediction-market band exposure accounting must use Redis WATCH/MULTI/EXEC, Redis scripting, or DB row locks, instead of current INCRBYFLOAT plus rollback.
3. Whether `cex-order-service/docs/RISK_CONTROL_RULES.md` becomes the canonical CEX risk policy source, superseding older service READMEs.
4. How to map runtime admin/DB/sys_config fields back to each governed risk control.

Source Freshness

Latest source checks were performed using workspace-scoped GitHub auth as thomson-ye.

Source	Latest Verified Ref
cex-order-service feature/main_fix	19dee65c6a640c9c78cc25a481f0a6a362b4347a
cex-order-service/ docs/ RISK_CONTROL_RULES.md blob	a802158211ee8761113ceb71cc3a0f892dbaa609
surfv2-dex-svm-keeper main	55baf70e520b8871a3168b94f997941487f25ea4
HQ risk wiki	Local draft under Turboflow-HQ/specs/risk- management/

Governance Structure

The HQ risk wiki is structured as:

File	Purpose
README.md	CTO dashboard and control view
policy-register.md	Central policy/control register
gap-register.md	Open gaps, severity, owner, decision needed
governance.md	Versioning, review cadence, approval gates
products/prediction-market.md	Prediction-market policy page
products/cex-risk-controls.md	CEX risk-control policy page
templates/policy-page- template.md	Template for future policy pages

Policy governance and enforcement are separated:

HQ policy wiki

- > product policy page
- > control register
- > implementation evidence
- > gap register
- > product repo changes / runtime config changes

Prediction-Market Risk

Policy source:

- Turboflow_Risk_Management_Spec_v3.0
- Prediction-market JSON config document
- surfv2-dex-svm-keeper implementation

Core controls:

- Settlement-band net exposure cap
- Signed UP/DOWN exposure offsetting
- Risk-weighted notional by duration and volatility regime
- Per-pair and special-period cap overrides
- Per-duration order min/max limits
- User/platform holding limits
- Rollback of risk accounting when downstream order creation fails

Current status: **Partially implemented.**

Key gaps:

Gap	Severity	Summary
RM-GAP-001	High	v3.0 requires independent volatility regimes per duration, but keeper currently uses the 30s cache for all durations.
RM-GAP-002	High	v3.0 requires atomic accounting via Redis WATCH/MULTI/EXEC or DB lock; keeper currently uses INCRBYFLOAT plus rollback.
RM-GAP-003	Medium	v3.0 duration/weight set differs from JSON config example.
RM-GAP-004	Medium	Low-volatility threshold differs: 0.10 vs 0.25.
RM-GAP-007	Low	Config doc says band_width_sec is 180, while JSON example and code default use 5.

CEX Risk Controls

Policy source:

- cex-order-service/docs/RISK_CONTROL_RULES.md on feature/main_fix

- Current cex-order-service code
- Older pair risk and max leverage service READMEs
- Oracle/slippage docs

Core controls:

- Tier Tag from Spread1k
- PM from Spread50k
- RM from Spread1k
- Profit Share eligibility
- Max leverage from spread, choppiness, funding rate, market hours, weekend mode
- MMR tiers and max order/holding limits
- Buffer rate and orderbook slippage
- Weekend and stock-close special modes

Current status: **Implemented with doc drift.**

Key gaps:

Gap	Severity	Summary
RM-GAP-005	Medium	max_leverage_service_README.md has stale spread-tier leverage values compared with current code and RISK_CONTROL_RULES.md.
RM-GAP-006	Low	RISK_CONTROL_RULES.md links to RISK_CONTROL_RULES_EN.md, but the branch does not contain that file.
RM-GAP-008	Medium	Admin/DB/sys_config edit surfaces for CEX and prediction risk controls are not fully mapped.
RM-GAP-009	Medium	Metrics, alerts, and runbooks are not fully inventoried.

Recommended Next Actions

1. Hold a CTO decision pass for RM-GAP-001 and RM-GAP-002; these are the highest-risk policy/implementation mismatches.
2. Promote RISK_CONTROL_RULES.md as canonical CEX policy, or explicitly mark it as branch-only until merged.
3. Create risk-config-surfaces.md to map each policy control to DB fields, sys_config keys, admin pages, owners, and release gates.
4. Add evidence columns for tests, metrics, alerts, and runtime config to each high-impact control.
5. Convert high and medium gaps into implementation tickets or explicit risk acceptances.

Current Readiness View

Capability	Current Readiness
Central policy wiki	Ready as draft
Product-level policy pages	Initial pages created
Version control model	Defined
Implementation status model	Defined
Gap tracking	Started
Runtime live status	Not yet connected
Metrics/alerts evidence	Not yet inventoried
Admin config map	Not yet inventoried

Conclusion

The governance foundation is now in place. Turboflow can centrally screen risk policy, ownership, implementation status, and gaps from HQ. The next maturity step is to connect policy controls to runtime config, tests, metrics, and alert evidence so the CTO view becomes a live operational control surface rather than a document-only register.